5-MINUTE REVISION

Design of Experiments

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 16 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- A well-designed experiment separates the effect of the treatments from other sources of variation, so that observed differences can be attributed to the treatments with confidence.
- Experimental error is the variation among experimental units that receive the same treatment; every design must provide a way of estimating it.
- The three basic principles of design, randomization, replication, and local control, work together: randomization protects against systematic allocation bias, replication permits estimation of experimental error and improves precision, and local control reduces experimental error by accounting for known sources of variation.
- An experimental unit is the smallest unit to which a treatment is independently applied; an observational unit is the unit on which the response is actually measured. Confusing the two leads to pseudo-replication.
- In a Randomized Block Design, similar experimental units are grouped into blocks, and all treatments are randomly assigned within each block, so that treatments are compared under similar conditions.

Quick revision · 2

- Precision can be further improved through good plot technique (keeping everything except the treatment uniform), excluding border plants from data collection (using only the net plot), and appropriate statistical analysis such as ANCOVA, which adjusts for covariates.
- R. A. Fisher developed the statistical foundations of experimental design at Rothamsted Experimental Station in the 1920s and 1930s, introducing ANOVA and, together with Frank Yates, the methodologies that remain central to agricultural research today.

Important concepts · 1

- Experiment → systematic procedure conducted under controlled conditions to discover an effect, test a hypothesis, or illustrate a known effect.
- Design of Experiments → statistical methodology for planning experiments so that treatment effects can be estimated accurately, efficiently, and without bias.
- Main objectives of DoE → validity, reliability, replicability, and optimality.

Important concepts · 2

- Treatment → any object of comparison in an experiment.
- Control → standard treatment used for comparison.
- Experimental unit → smallest unit to which a treatment is independently applied.

Important concepts · 3

- Observation unit → unit on which the response is measured.
- Response → measurable outcome observed after treatment application.
- Factor → variable whose influence on a response is being studied.

Important concepts · 4

- Factor level → value or setting taken by a factor.
- Experimental error → variation in response caused by extraneous factors among experimental units receiving the same treatment.
- Extraneous factors → factors other than the treatments that cause variation in the response.

Important concepts · 5

- Randomization → assignment of treatments to experimental units purely by chance.
- Complete randomization → treatments are assigned without blocking restrictions, subject to the pre-specified treatment counts; allocation may be equal or unequal.
- Replication → application of the same treatment to more than one experimental unit.

Important concepts · 6

- Local control → grouping similar experimental units into blocks before assigning treatments.
- Blocking → another term for local control.
- Basic principles of design → randomization, replication, and local control.

Important concepts · 7

- Randomization → protects against allocation bias and balances unknown sources of variation in expectation; a particular allocation can remain unbalanced.
- Replication → improves reliability and precision and permits estimation of experimental error.
- Local control → reduces experimental error by accounting for known sources of variation.

Important concepts · 8

- Randomized Block Design → design in which similar experimental units are grouped into blocks and all treatments are randomly assigned within each block.
- Border effect → influence of neighbouring plots on plants located near plot borders.
- Net plot → inner portion of a plot from which observations are recorded after excluding border plants.

Important concepts · 9

- Proper plot technique → keeping all management practices other than the treatments as uniform as possible.
- ANCOVA → statistical method that adjusts the response using one or more covariates to improve precision.
- Good experimental design → supports unbiased, precise and interpretable comparisons, uses resources efficiently, and defines the conditions to which conclusions apply.

Self-check from the chapter

1. The condition or combination of conditions applied to an experimental unit is called a _____. **[Treatment]**
2. A standard treatment used for comparison is called a _____. **[Control]**
3. The smallest unit to which a treatment is independently assigned is the _____ unit. **[Experimental]**
4. A variable whose levels are investigated in an experiment is called a _____. **[Factor]**